SECTION 1 — MODEL DESCRIPTION

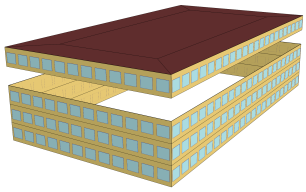
BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	Mid-rise (4-9 storeys)
Vintage	1945-1983

HVAC SYSTEM

Heating	Electric boiler (hot water) + Hydronic baseboards / radiators
Cooling	Water-cooled chiller
Air distribution	Constant air volume (CAV) (1 air loop)
Heat recovery	None

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	9,685
Conditioned floor area [m²]	9,685
Above-ground storeys	5
Climate zone	7
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	40.0
North	40.0
East	40.0
South	40.0
West	40.0

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.43
Roof [W/m²K]	0.22
Exposed floors / slabs [W/m²K]	2.94
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	5.00
SHGC	0.80

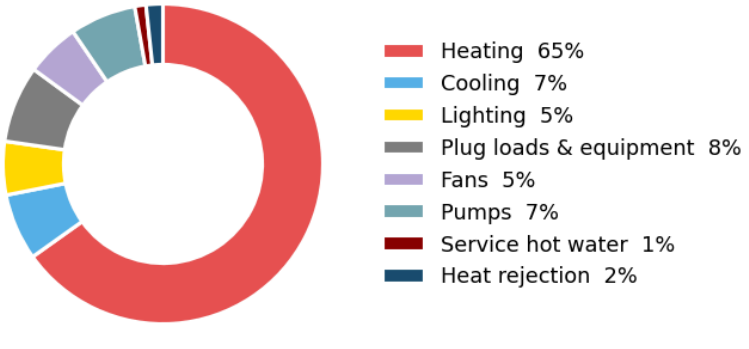
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	19.5
Lighting power density [W/m²]	6.7
Equipment power density [W/m²]	7.5

ANNUAL END USES [kWh/year]

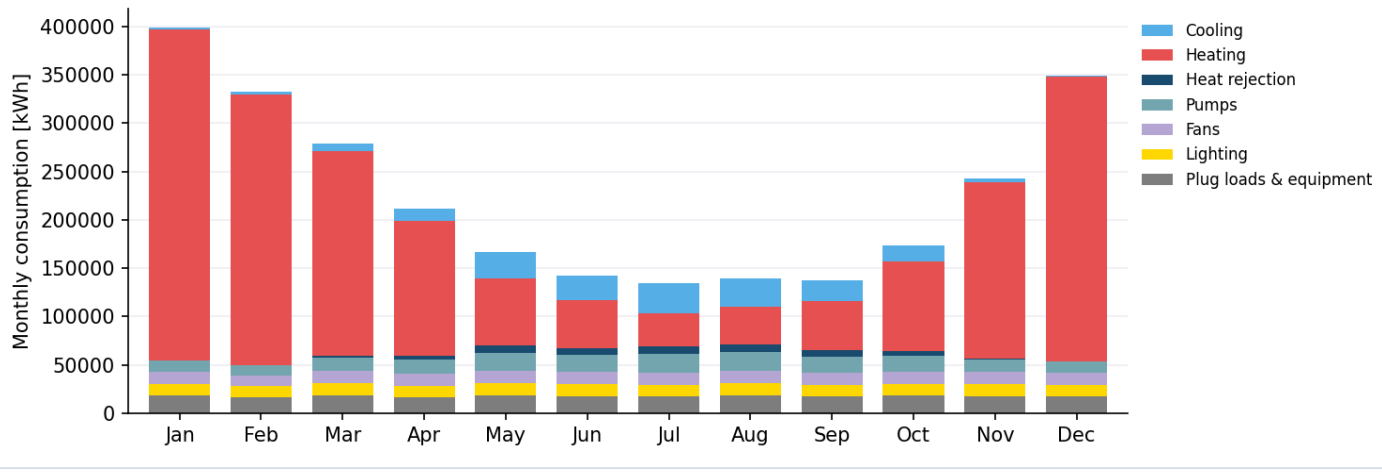
END USE	ELECTRICITY	DISTRICT HEATING	TOTAL	kWh/m²
Heating	1,783,931	0	1,783,931	184.2
Cooling	184,264	0	184,264	19.0
Interior lighting	147,814	0	147,814	15.3
Plug loads & equipment	213,039	0	213,039	22.0
Fans	148,903	0	148,903	15.4
Pumps	182,853	0	182,853	18.9
Heat rejection	46,442	0	46,442	4.8
Service hot water	0	31,075	31,075	3.2
Total			**2,738,320**	**282.7**

SHARE OF ANNUAL CONSUMPTION



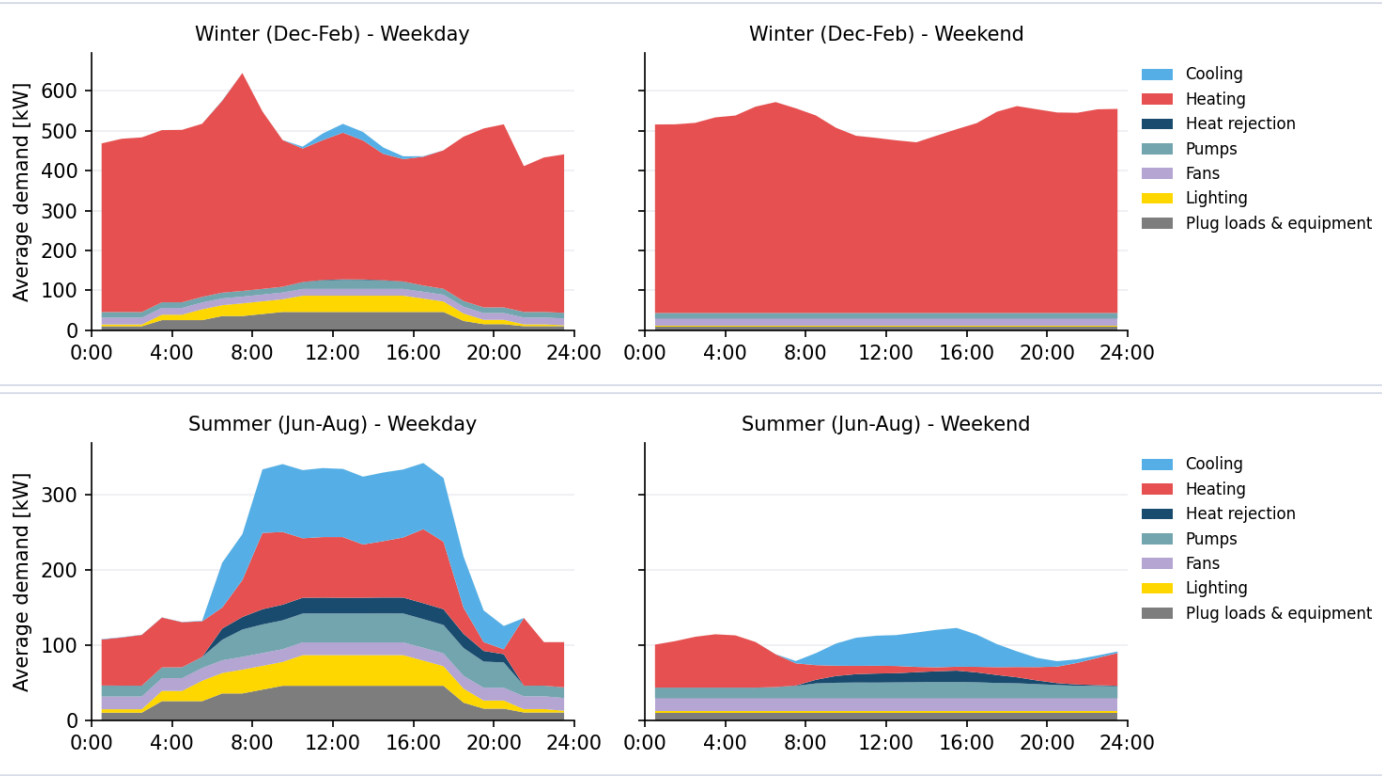
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	OTHER FUELS	PEAK ELECTRIC DEMAND
283 kWh/m²/yr (total area)	2,738,320 kWh/yr	2,707,245 kWh/yr	0 kWh/yr	31,075 kWh/yr	2,029 kW · 209.5 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (9,685 m²). Offices are not normalized per unit.
- Archetype: Mid-rise office (4-9 storeys), built 1945-1983.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 24-DEC-05:10.